

In Man Kang

associate professor

基本信息

姓名: In Man Kang

所属院校: 庆北国立大学

地区: South Korea

年龄范围: 40岁至50岁

企业经历: 有

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

射频集成电路

薄膜材料与表面物理

模拟与射频IC设计

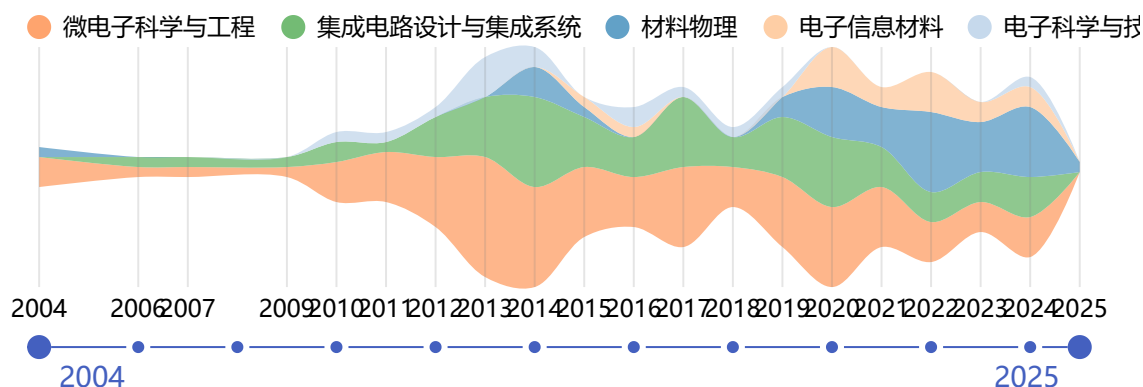
低功耗电路与能量管理

半导体与光电子材料

个人简介

In Man Kang (ORCID: 0000 - 0002 - 7726 - 9740) is an associate professor at the School of Electronics Engineering, Kyungpook National University. He obtained his B.S. from Kyungpook National University in 2001 and Ph.D. from Seoul National University in 2007. He was a teaching assistant at SNU from 2001 - 2006 and a senior engineer at Samsung from 2007 - 2010. His research interests include CMOS RF modelling, silicon nanowire devices, etc. He is a member of IEEE EDS and has co - authored multiple journal articles.

近年发表的论文



Design and analysis of quasi-vertical multi-fin GaN power devices based on epitaxially grown GaN-on-sapphire

JOURNAL OF SCIENCE-ADVANCED MATERIALS AND DEVICES

Hong, Jeong Woo; Lee, Sang Ho; Park, Jin; Kim, Min Seok; Bae, Seung Ji; Koh, Won Suk; San Yun, Gang; Kang, In Man

被引用次数: 0

2025

Enhancing the nonvolatile properties of sol-gel-processed Y2O3 RRAM devices by suppressing oxygen vacancy formation

MATERIALS SCIENCE IN SEMICONDUCTOR PROCESSING

Lee, Sangwoo; Cho, Yoonjin; Heo, Seongwon; Choi, Suhyeon; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Lee, Won-Yong; Jang, Jaewon

被引用次数: 0

2025

Simulation of capacitorless DRAM based on polycrystalline silicon with a vertical underlap structure and a separated channel layer

JAPANESE JOURNAL OF APPLIED PHYSICS

Bae, Seung Ji; Lee, Sang Ho; Park, Jin; Kang, Ga Eon; Heo, Jun Hyeok; Jeon, So Ra; Kim, Min Seok; Hong, Jeong Woo; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 1

2024

Bulk fin-type field-effect transistor-based capacitorless dynamic random-access memory with strong resistance to geometrical variations

JAPANESE JOURNAL OF APPLIED PHYSICS

Kim, Min Seok; Lee, Sang Ho; Park, Jin; Kang, Ga Eon; Heo, Jun Hyeok; Jeon, So Ra; Bae, Seung Ji; Hong, Jeong Woo; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 1

2024

Formation of Cluster-Structured Metallic Filaments in Organic Memristors for Wearable Neuromorphic Systems with Bio-Mimetic Synaptic Weight Distributions

ADVANCED SCIENCE

Jung, Uihoon; Kim, Miseong; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man; Lee, Sin-Hyung

被引用次数: 12

2024

Verifying the physical role of upper-active-layer on charge transport together with bias stability in bilayer-channel oxide thin-film transistors

SURFACES AND INTERFACES

Lee, Jinuk; Eun, Jun-Su; Na, Jeong-Hyeon; Park, Won; Park, Jun-Hyeong; Feng, Junhao; Jang, Jaewon; Kang, In Man; Park, Jaehoon; Zhang, Xue; Kim, Do-Kyung; Bae, Jin-Hyuk

被引用次数: 0

2024

Complementary resistive switching characteristics of solid electrolyte chalcogenide AgxTe nanoparticles for high-density crossbar random access memory

MATERIALS TODAY NANO

Lee, Won-Yong; Cho, Yoonjin; Lee, Sangwoo; Kim, Kwangeun; Bae, Jin-Hyuk; Kang, In-Man; Jang, Jaewon

被引用次数: 0

2024

Viable Approach of Tuning Oxide Semiconductor Thin Films in Solution-Processed Heterojunction Thin Films Transistors for Both Higher Performances and Stability

ADVANCED ELECTRONIC MATERIALS

Eun, Jun-Su; Lee, Jinuk; Na, Jeong-Hyeon; Park, Jun-Hyeong; Park, Won; Feng, Junhao; Seo, Kyung-Ho; Jang, Jaewon; Kang, In Man; Kim, Do-Kyung; Bae, Jin-Hyuk

被引用次数: 0

2024

Statistical Analysis of Increased Immunity to Poly-Si Grain Boundaries in Nanosheet CMOS Logic Inverter Through Sheet Stacking

SILICON

Kim, Min Seok; Lee, Sang Ho; Park, Jin; Jeon, So Ra; Bae, Seung Ji; Hong, Jeong Woo; Jang, Jaewon; Bae, Jin-Hyuk; Yoon, Young Jun; Kang, In Man

被引用次数: 0

2024

Improving the Nonvolatile Memory Characteristics of Sol-Gel-Processed Y2O3 RRAM Devices Using Mono-Ethanolamine Additives

MATERIALS

Heo, Seongwon; Choi, Soohyun; Lee, Sangwoo; Cho, Yoonjin; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Lee, Won-Yong; Jang, Jaewon

被引用次数: 0

2024

UV/Ozone-Treated and Sol-Gel-Processed Y2O3 Insulators Prepared Using Gelation-Delaying Precursors

NANOMATERIALS

Lee, Sangwoo; Cho, Yoonjin; Heo, Seongwon; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Lee, Won-Yong; Jang, Jaewon

被引用次数: 0

2024

Effect of Electrochemically Active Top Electrode Materials on Nanoionic Conductive Bridge Y2O3 Random-Access Memory

NANOMATERIALS

Cho, Yoonjin; Lee, Sangwoo; Heo, Seongwon; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Lee, Won-Yong; Jang, Jaewon

被引用次数: 0

2024

Dependence of Positive Bias Stress Instability on Threshold Voltage and Its Origin in Solution-Processed Aluminum-Doped Indium Oxide Thin-Film Transistors

NANOMATERIALS

Na, Jeong-Hyeon; Park, Jun-Hyeong; Park, Won; Feng, Junhao; Eun, Jun-Su; Lee, Jinuk; Lee, Sin-Hyung; Jang, Jaewon; Kang, In Man; Kim, Do-Kyung; Bae, Jin-Hyuk

被引用次数: 1

2024

Capacitorless One-Transistor Dynamic Random-Access Memory with Novel Mechanism: Self-Refreshing

NANOMATERIALS

Lee, Sang Ho; Park, Jin; Yoon, Young Jun; Kang, In Man

被引用次数: 1

2024

One-Stop Strategy for Obtaining Controllable Sensitivity and Feasible Self-Patterning in Silver Nanowires/Elastomer Nanocomposite-Based Stretchable Ultrathin Strain Sensors

BIOMACROMOLECULES

Lee, Jinuk; Park, Jun-Ik; Lee, Sin-Hyung; Jang, Jaewon; Kang, In-Man; Park, Jaehoon; Zhang, Xue; Kim, Do-Kyung; Bae, Jin-Hyuk

被引用次数: 1

2023

Bulk-fin field-effect transistor-based capacitorless dynamic random-access memory and its immunity to the work-function variation effect

JAPANESE JOURNAL OF APPLIED PHYSICS

Lee, Sang Ho; Park, Jin; Kim, Geon Uk; Kang, Ga Eon; Heo, Jun Hyeok; Jeon, So Ra; Yoon, Young Jun; Seo, Jae Hwa; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 4

2023

Organic Memristor-Based Flexible Neural Networks with Bio-Realistic Synaptic Plasticity for Complex Combinatorial Optimization

ADVANCED SCIENCE

Kim, Hyeongwook; Kim, Miseong; Lee, Aejin; Park, Hea-Lim; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man; Kim, Eun-Sol; Lee, Sin-Hyung

被引用次数: 18

2023

Improved Negative Bias Stability of Sol-Gel-Processed SnO₂ Thin-Film Transistors with Vertically Controlled Carrier Concentrations

ACS APPLIED ELECTRONIC MATERIALS

Lee, Taehun; Kim, Kyoungdu; Kim, Hae-In; Lee, Sin-Hyung; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Lee, Won-Yong; Jang, Jaewon

被引用次数: 5

2023

Sol-Gel-Processed Y₂O₃ Multilevel Resistive Random-Access Memory Cells for Neural Networks

NANOMATERIALS

Lee, Taehun; Kim, Hae-In; Cho, Yoonjin; Lee, Sangwoo; Lee, Won-Yong; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Lee, Sin-Hyung; Jang, Jaewon

Low-Temperature Enhancement-Mode Amorphous Oxide Thin-Film Transistors in Solution Process Using a Low-Pressure Annealing

NANOMATERIALS

Park, Won; Park, Jun-Hyeong; Eun, Jun-Su; Lee, Jinuk; Na, Jeong-Hyeon; Lee, Sin-Hyung; Jang, Jaewon; Kang, In Man; Kim, Do-Kyung; Bae, Jin-Hyuk

被引用次数: 3

2023

Simulation of Capacitorless DRAM Based on the Polycrystalline Silicon Nanotube Structure with Multiple Grain Boundaries

NANOMATERIALS

Park, Jin; Lee, Sang-Ho; Kang, Ga-Eon; Heo, Jun-Hyeok; Jeon, So-Ra; Kim, Min-Seok; Bae, Seung-Ji; Hong, Jeong-Woo; Jang, Jae-won; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In-Man

被引用次数: 5

2023

Improvement in Switching Characteristics and Bias Stability of Solution-Processed Zinc-Tin Oxide Thin Film Transistors via Simple Low-Pressure Thermal Annealing Treatment

NANOMATERIALS

Feng, Junhao; Jeon, Sang-Hwa; Park, Jaehoon; Lee, Sin-Hyung; Jang, Jaewon; Kang, In Man; Kim, Do-Kyung; Bae, Jin-Hyuk

被引用次数: 0

2023

Thickness dependence of resistive switching characteristics of the sol-gel processed Y2O3 RRAM devices

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Kim, Kyoungdu; Kim, Hae-In; Lee, Taehun; Lee, Won-Yong; Bae, Jin-Hyuk; Kang, In Man; Lee, Sin-Hyung; Kim, Kwangeun; Jang, Jaewon

被引用次数: 5

2023

Electrical Performance Depending on the Grain Boundary-location in the Multiple Nanosheet Tunneling Field-effect Transistor based on the Poly-Si

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kang, Ga Eon; Lee, Sang Ho; Park, Jin; Min, So Ra; Kim, Geon Uk; Heo, Jun Hyeok; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 0

2023

Analysis of Multiple Fin-type Vertical GaN Power Transistors based on Bulk GaN Substrates

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Heo, Jun Hyeok; Lee, Sang Ho; Park, Jin; Min, So Ra; Kim, Geon Uk; Kang, Ga Eon; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 1

2023

Design of a Vertical Cylinder GaN Junctionless FET based on its GaN-on-GaN Substrate and Electrical Performance

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Jeon, So Ra; Lee, Sang Ho; Park, Jin; Kang, Ga Eon; Heo, Jun Hyeok; Kim, Min Seok; Bae, Seung Ji; Hong, Jeong Woo; Kang, In Man

被引用次数: 0

2023

Viable strategy to minimize trap states of patterned oxide thin films for both exceptional electrical performance and uniformity in sol-gel processed transistors

CHEMICAL ENGINEERING JOURNAL

Kim, Do-Kyung; Seo, Kyeong-Ho; Kwon, Dae-Hyeon; Jeon, Sang-Hwa; Hwang, Yu-Jin; Wang, Ziyuan; Park, Jaehoon; Lee, Sin-Hyung; Jang, Jaewon; Kang, In Man; Zhang, Xue; Bae, Jin-Hyuk

被引用次数: 8

2022

Simulation of CMOS logic inverter based on vertically stacked polycrystalline silicon nanosheet gate-all-around MOSFET and its electrical characteristics

CURRENT APPLIED PHYSICS

Min, So Ra; Lee, Sang Ho; Park, Jin; Kim, Geon Uk; Kang, Ga Eon; Heo, Jun Hyeok; Yoon, Young Jun; Seo, Jae Hwa; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 1

2022

Organizing Reliable Polymer Electrode Lines in Flexible Neural Networks via Coffee Ring-Free Micromolding in Capillaries

ACS APPLIED MATERIALS & INTERFACES

Park, Min Woo; Kim, Do-Yun; An, Ukju; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man; Lee, Sin-Hyung

被引用次数: 17

2022

Environmentally and Electrically Stable Sol-Gel-Deposited SnO₂ Thin-Film Transistors with Controlled Passivation Layer Diffusion Penetration Depth That Minimizes Mobility Degradation

ACS APPLIED MATERIALS & INTERFACES

Lee, Won-Yong; Kim, Do Won; Kim, Hyeon Joong; Kim, Kyoungdu; Lee, Sin-Hyung; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Jang, Jaewon

被引用次数: 14

2022

Physico-Chemical Origins of Electrical Characteristics and Instabilities in Solution-Processed ZnSnO Thin-Film Transistors

COATINGS

Wang, Ziyuan; Jeon, Sang-Hwa; Hwang, Yu-Jin; Lee, Sin-Hyung; Jang, Jaewon; Kang, In Man; Kim, Do-Kyung; Bae, Jin-Hyuk

被引用次数: 2

2022

Design of a Capacitorless DRAM Based on a Polycrystalline-Silicon Dual-Gate MOSFET with a Fin-Shaped Structure

NANOMATERIALS

An, Hee Dae; Lee, Sang Ho; Park, Jin; Min, So Ra; Kim, Geon Uk; Yoon, Young Jun; Seo, Jae Hwa; Cho, Min Su; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 5

2022

Importance of Structural Relaxation on the Electrical Characteristics and Bias Stability of Solution-Processed ZnSnO Thin-Film Transistors

NANOMATERIALS

Hwang, Yu-Jin; Kim, Do-Kyung; Jeon, Sang-Hwa; Wang, Ziyuan; Park, Jaehoon; Lee, Sin-Hyung; Jang, Jaewon; Kang, In Man; Bae, Jin-Hyuk

被引用次数: 4

2022

Improved Environment Stability of Y2O3 RRAM Devices with Au Passivated Ag Top Electrodes

MATERIALS

Kim, Hae-In; Lee, Taehun; Lee, Won-Yong; Kim, Kyoungdu; Bae, Jin-Hyuk; Kang, In-Man; Lee, Sin-Hyung; Kim, Kwangeun; Jang, Jaewon

被引用次数: 5

2022

Enhanced Switching Reliability of Sol-Gel-Processed Y2O3 RRAM Devices Based on Y2O3 Surface Roughness-Induced Local Electric Field

MATERIALS

Kim, Do-Won; Kim, Hyeon-Joong; Lee, Won-Yong; Kim, Kyoungdu; Lee, Sin-Hyung; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Jang, Jaewon

被引用次数: 15

2022

Flexible Sol-Gel-Processed Y2O3 RRAM Devices Obtained via UV/Ozone-Assisted Photochemical Annealing Process

MATERIALS

Kim, Hyeon-Joong; Kim, Do-Won; Lee, Won-Yong; Kim, Kyoungdu; Lee, Sin-Hyung; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Jang, Jaewon

被引用次数: 10

2022

Analysis for DC and RF Characteristics Recessed-Gate GaN MOSFET Using Stacked TiO2/Si3N4 Dual-Layer Insulator

MATERIALS

Min, So-Ra; Cho, Min-Su; Lee, Sang-Ho; Park, Jin; An, Hee-Dae; Kim, Geon-Uk; Yoon, Young-Jun; Seo, Jae-Hwa; Jang, Jae-Won; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In-Man

被引用次数: 4

2022

Systematic Engineering of Metal Ion Injection in Memristors for Complex Neuromorphic Computing with High Energy Efficiency

ADVANCED INTELLIGENT SYSTEMS

Kim, Seong Eun; Kim, Min-Hwi; Jang, Jisu; Kim, Hyungjin; Kim, Sungjun; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man; Lee, Sin-Hyung

被引用次数: 17

2022

Room-Temperature High-Detectivity Flexible Near-Infrared Photodetectors with Chalcogenide Silver Telluride Nanoparticles

ACS OMEGA

Lee, Won-Yong; Kim, Kyoungdu; Lee, Sin-Hyung; Bae, Jin-Hyuk; Kang, In-Man; Park, Minsu; Kim, Kwangeun; Jang, Jaewon

Design of a Capacitorless DRAM Based on Storage Layer Separated Using Separation Oxide and Polycrystalline Silicon

ELECTRONICS

Kim, Geon Uk; Yoon, Young Jun; Seo, Jae Hwa; Lee, Sang Ho; Park, Jin; Kang, Ga Eon; Heo, Jun Hyeok; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 3

2022

Enhanced switching ratio of sol-gel-processed Y₂O₃ RRAM device by suppressing oxygen vacancy formation at high annealing temperatures

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Kim, Kyoungdu; Lee, Changmin; Lee, Won-Yong; Kim, Do Won; Kim, Hyeon Joong; Lee, Sin-Hyung; Bae, Jin-Hyuk; Kang, In-Man; Jang, Jaewon

被引用次数: 10

2022

Analysis and Optimization for Characteristics of Vertical GaN Junctionless MOSFETs Depending on Specifications of GaN Substrates

JOURNAL OF ELECTRICAL ENGINEERING & TECHNOLOGY

An, Hee Dae; Lee, Sang Ho; Park, Jin; Min, So Ra; Kim, Geon Uk; Yoon, Young Jun; Seo, Jae Hwa; Cho, Min Su; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 3

2022

Fabrication and Performances of Recessed Gate AlGa_N/Ga_N MOSFETs with Si₃N₄/TiO₂ Stacked Dual Gate Dielectric

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

An, Hee Dae; Min, So Ra; Lee, Sang Ho; Park, Jin; Kim, Geon Uk; Yoon, Young Jun; Seo, Jae Hwa; Cho, Min Su; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 2

2022

3-D stacked polycrystalline-silicon-MOSFET-based capacitorless DRAM with superior immunity to grain-boundary's influence

SCIENTIFIC REPORTS

Lee, Sang Ho; Park, Jin; Min, So Ra; Kim, Geon Uk; Jang, Jaewon; Bae, Jin-Hyuk; Lee, Sin-Hyung; Kang, In Man

被引用次数: 3

2022

Impact of process-dependent Si_Nx passivation on proton-induced degradation in Ga_N MIS-HEMTs

RESULTS IN PHYSICS

Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Lee, Eun Je; Kim, Dong-Seok

被引用次数: 5

2021

Fluoropolymer-based organic memristor with multifunctionality for flexible neural network system

NPJ FLEXIBLE ELECTRONICS

Kim, Min-Hwi; Park, Hea-Lim; Kim, Min-Hoi; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man; Lee, Sin-Hyung

Extremely bias stress stable enhancement mode sol-gel-processed SnO₂ thin-film transistors with Y₂O₃ passivation layers

APPLIED SURFACE SCIENCE

Lee, Changmin; Lee, Won-Yong; Kim, Do Won; Kim, Hyeon Joong; Bae, Jin-Hyuk; Kang, In-Man; Lim, Doohyeok; Kim, Kwangeun; Jang, Jaewon

被引用次数: 23

2021

Improving of Sensitivity of PbS Quantum Dot Based SWIR Photodetector Using P3HT

MATERIALS

Seo, Kyeong-Ho; Jang, Jaewon; Kang, In Man; Bae, Jin-Hyuk

被引用次数: 4

2021

Single-event transient characteristics of vertical gate-all-around junctionless field-effect transistor on bulk substrate

APPLIED PHYSICS A-MATERIALS SCIENCE & PROCESSING

Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Kim, Dong-Seok

被引用次数: 7

2021

Design of Capacitorless DRAM Based on Polycrystalline Silicon Nanotube Structure

IEEE ACCESS

Park, Jin; Cho, Min Su; Lee, Sang Ho; An, Hee Dae; Min, So Ra; Kim, Geon Uk; Yoon, Young Jun; Seo, Jae Hwa; Lee, Sin-Hyung; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man

被引用次数: 9

2021

Polycrystalline-Silicon-MOSFET-Based Capacitorless DRAM With Grain Boundaries and Its Performances

IEEE ACCESS

Lee, Sang Ho; Jang, Won Douk; Yoon, Young Jun; Seo, Jae Hwa; Mun, Hye Jin; Cho, Min Su; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man

被引用次数: 11

2021

Design optimization of GaN diode with p-GaN multi-well structure for high-efficiency betavoltaic cell

NUCLEAR ENGINEERING AND TECHNOLOGY

Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Lee, Jung-Hee; Kim, Dong-Seok

被引用次数: 3

2021

Improved Negative Bias Stress Stability of Sol-Gel-Processed Li-Doped SnO₂ Thin-Film Transistors

ELECTRONICS

Kim, Hyeon-Joong; Kim, Do-Won; Lee, Won-Yong; Lee, Sin-Hyung; Bae, Jin-Hyuk; Kang, In-Man; Jang, Jaewon

被引用次数: 7

2021

Influence of Active Channel Layer Thickness on SnO₂ Thin-Film Transistor Performance

ELECTRONICS

Kim, Do Won; Kim, Hyeon Joong; Lee, Changmin; Kim, Kyoungdu; Bae, Jin-Hyuk; Kang, In-Man; Jang, Jaewon

被引用次数: 11

2021

Experimental and simulation study of power performance improvement of GaN PIN betavoltaic cell

INTERNATIONAL JOURNAL OF ENERGY RESEARCH

Kim, Dong-Seok; Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Lee, Jung-Hee

被引用次数: 7

2021

Design and optimization of GaN-based betavoltaic cell for enhanced output power density

INTERNATIONAL JOURNAL OF ENERGY RESEARCH

Yoon, Young Jun; Lee, Jae Sang; Kang, In Man; Lee, Jung-Hee; Kim, Dong-Seok

被引用次数: 8

2021

Effects of Proton Irradiation on the Current Characteristics of SiN-Passivated AlGaIn/GaN MIS-HEMTs Using a TMAH-Based Surface Pre-Treatment

MICROMACHINES

Yoon, Young Jun; Lee, Jae Sang; Suk, Jae Kwon; Kang, In Man; Lee, Jung Hee; Lee, Eun Je; Kim, Dong Seok

被引用次数: 7

2021

Effect of Work-function Variation on Transfer Characteristics and Memory Performances for Gate-all-around JLFET based Capacitorless DRAM

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Sang Ho; Yoon, Young Jun; Seo, Jae Hwa; Cho, Min Su; Park, Jin; An, Hee Dae; Min, So Ra; Kim, Geon Uk; Kang, In Man

被引用次数: 1

2021

Design and Analysis of DC/DC Boost Converter using Vertical GaN Power Device based on Epitaxially Grown GaN-on-sapphire

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Cho, Min Su; Lee, Sang Ho; An, Hee Dae; Park, Jin; Min, So Ra; Kim, Geon Uk; Yoon, Young Jun; Seo, Jae Hwa; Kang, In Man

被引用次数: 0

2021

Electrical Performances of GaN-based Vertical Trench MOSFETs with Cylindrical and Hexagonal Structure

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Geon Uk; Yoon, Young Jun; Seo, Jae Hwa; Cho, Min Su; Lee, Sang Ho; Park, Jin; An, Hee Dae; Min, So Ra; Kang, In Man

被引用次数: 0

2021

Polycrystalline silicon metal-oxide-semiconductor field-effect transistor-based stacked multi-layer one-transistor dynamic random-access memory with double-gate structure for the embedded systems

JAPANESE JOURNAL OF APPLIED PHYSICS

Jang, Won Douk; Yoon, Young Jun; Cho, Min Su; Jung, Jun Hyeok; Lee, Sang Ho; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man

被引用次数: 5

2020

Control of silver nanowire-elastomer nanocomposite networks through elaborate direct printing for ultrathin and stretchable strain sensors

COMPOSITES SCIENCE AND TECHNOLOGY

Park, Jun-Ik; Kim, Do-Kyung; Jang, Jaewon; Kang, In Man; Kim, Hyeok; Park, Jaehoon; Nam, Il Woo; Lang, Philippe; Bae, Jin-Hyuk

被引用次数: 11

2020

Expeditious and eco-friendly solution-free self-patterning of sol-gel oxide semiconductor thin films

MATERIALS & DESIGN

Kim, Do-Kyung; Park, Jun-Ik; Jang, Jaewon; Kang, In Man; Park, Jaehoon; Bae, Jin-Hyuk

被引用次数: 8

2020

Contact line curvature-induced molecular misorientation of a surface energy patterned organic semiconductor in meniscus-guided coating

APPLIED SURFACE SCIENCE

Kim, Do-Kyung; Vincent, Premkumar; Jang, Jaewon; Kang, In Man; Kim, Hyeok; Lang, Philippe; Choi, Muhan; Bae, Jin-Hyuk

被引用次数: 11

2020

Simulation of capacitorless dynamic random access memory based on junctionless FinFETs using grain boundary of polycrystalline silicon

APPLIED PHYSICS A-MATERIALS SCIENCE & PROCESSING

Cho, Min Su; Mun, Hye Jin; Lee, Sang Ho; Jang, Jaewon; Bae, Jin-Hyuk; Kang, In Man

被引用次数: 5

2020

Fabrication of AlGaIn/GaN MISHEMT with dual-metal gate electrode and its performances

APPLIED PHYSICS A-MATERIALS SCIENCE & PROCESSING

Jung, Jun Hyeok; Cho, Min Su; Jang, Won Douk; Lee, Sang Ho; Jang, Jaewon; Bae, Jin-Hyuk; Yoon, Young Jun; Kang, In Man

被引用次数: 14

2020

Fabrication of AlGaIn/GaN Fin-Type HEMT Using a Novel T-Gate Process for Improved Radio-Frequency Performance

IEEE ACCESS

Cho, Min Su; Seo, Jae Hwa; Lee, Sang Ho; Jang, Hwan Soo; Kang, In Man

被引用次数: 13

2020

Enhancement Mode Flexible SnO₂ Thin Film Transistors Via a UV/Ozone-Assisted Sol-Gel Approach

IEEE ACCESS

Jang, Bongho; Kang, Hongki; Lee, Won-Yong; Bae, Jin-Hyuk; Kang, In-Man; Kim, Kwangeun; Kwon, Hyuk-Jun; Jang, Jaewon

Effects of Contact Potential and Sidewall Surface Plane on the Performance of GaN Vertical Nanowire MOSFETs for Low-Voltage Operation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Son, Dong-Hyeok; Thingujam, Terirama; Kim, Jeong-Gil; Kim, Dae-Hyun; Kang, In Man; Im, Ki-Sik; Theodorou, Christoforos; Ghibaudo, Gerard; Cristoloveanu, Sorin; Lee, Jung-Hee

被引用次数: 7

2020

One-Transistor Dynamic Random-Access Memory Based on Gate-All-Around Junction-Less Field-Effect Transistor with a Si/SiGe Heterostructure

ELECTRONICS

Yoon, Young Jun; Lee, Jae Sang; Kim, Dong-Seok; Lee, Sang Ho; Kang, In Man

被引用次数: 12

2020

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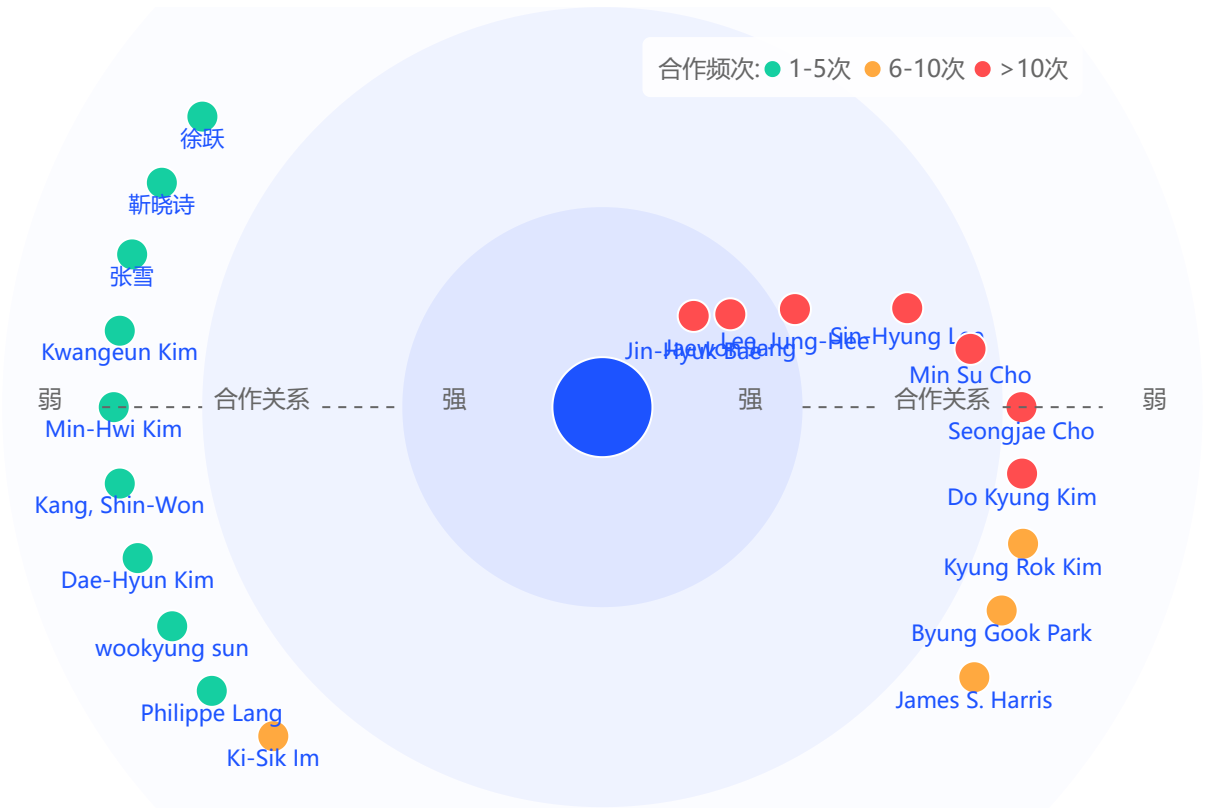
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